

Moulay Ismail University
National School of Arts and Crafts

Department of Industrial Engineering and Production
Master's Thesis Defense

Project Title :

**Community Detection in Social Networks for Targeted Marketing and
Recommender Systems**

Presented by:
MOUDDEN Hamza

Supervised by:
Pr.ELYAZID Akachar

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Presentation Outline

01 Introduction & Problem Statement

02 Background & Key Concepts

03 Literature Review

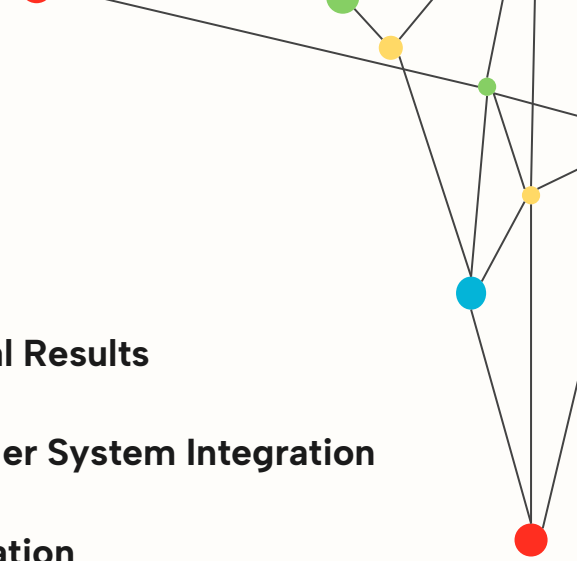
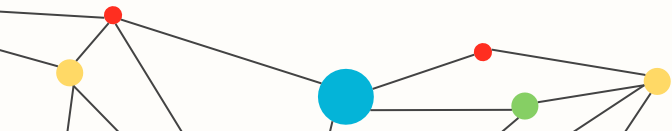
04 Methodology

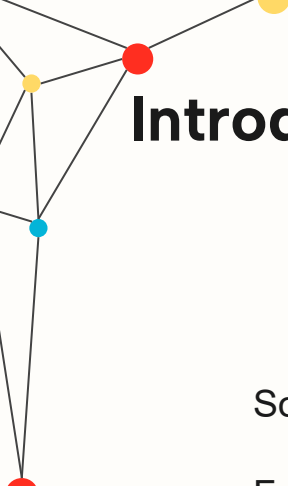
05 Experimental Results

06 Recommender System Integration

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08 Conclusion & Future Perspectives





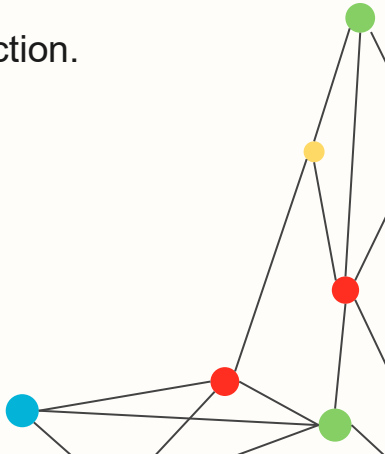
Introduction

Social media platforms generate massive networks of user interactions.

Facebook: 2+ billion active users.

Interactions include likes, shares, comments, follows.

These interactions form intricate networks revealing patterns of human connection.





Problem Statement

Despite numerous community detection algorithms available, selecting the right one remains challenging:

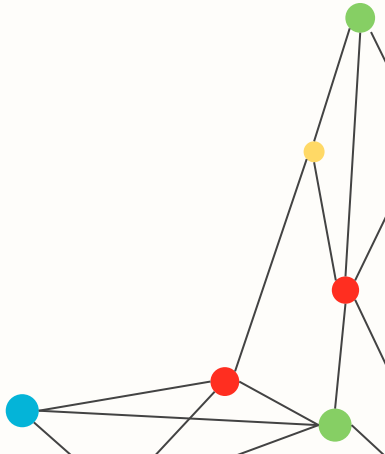
1. **Algorithmic Diversity** Different principles (modularity optimization, random walks, deep learning).
2. **Evaluation Gap** Most studies use modularity that don't predict practical usefulness.
3. **Limited Comparison** Modern vs. classical methods not systematically compared, especially at scale.
4. **Practical Utility** Lack of unified evaluation framework for real-world applications.

Research Question: *How do different community detection approaches compare in effectiveness and practical utility for recommender systems?*





Research Objectives

1. Study and categorize community detection approaches (classical & modern)
 2. Implement representative algorithms from different categories
 3. Apply algorithms to real-world and synthetic network datasets
 4. Evaluate using structural metrics (modularity, NMI, ARI)
 5. Assess practical usefulness for targeted marketing & recommendation
 6. Compare performance and identify strengths/limitations
 7. Provide insights on trade-offs between simplicity, scalability, and effectiveness
 8. Build an interactive web application for illustration.
- 



Background – Community Detection

What is a Community?

Definition: Groups of nodes with denser internal connections than external connections

Community Types:

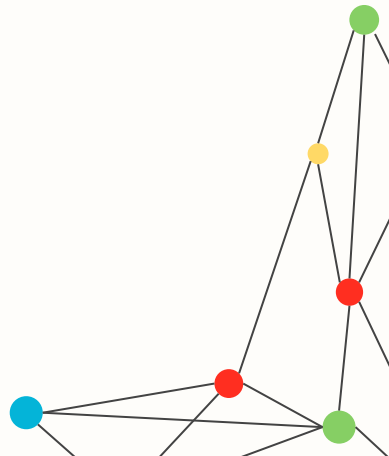
- **Disjoint:** Each node belongs to exactly one community
- **Overlapping:** Nodes can belong to multiple communities (clique percolation)

Modularity (Q): Compares internal edges vs expected in random graph

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

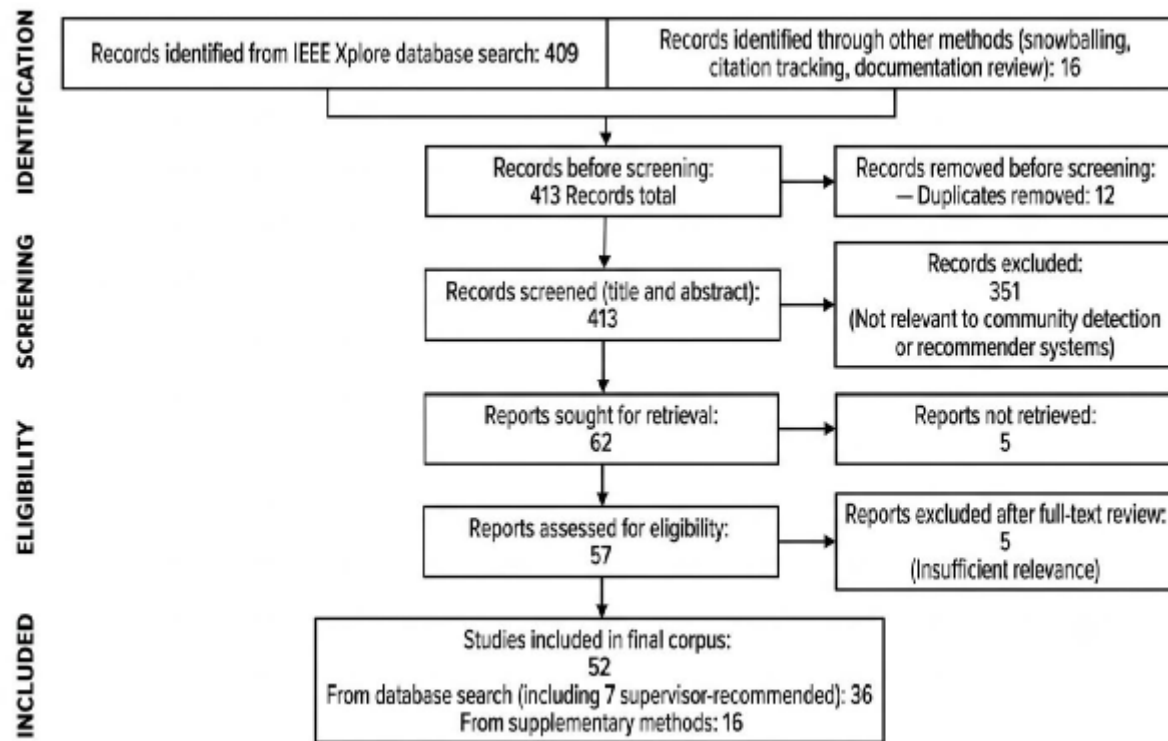
Interpretation: High Q = strong community structure

Limitation: Resolution limit for small communities (Fortunato, 2010)



Literature Review – Methodology

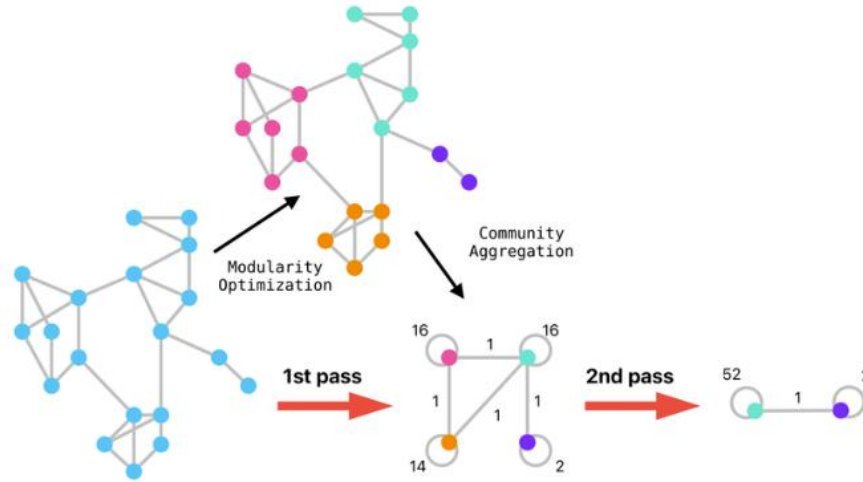
Prisma flow diagram



Literature Review – Classical Algorithms

Modularity-Based Approaches

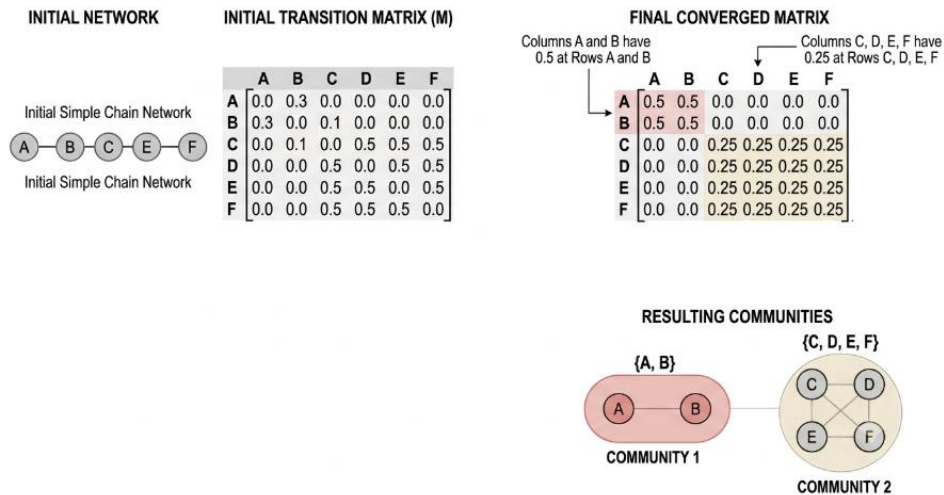
<u>Algorithm</u>	<u>What It Does</u>	<u>Limitation</u>
Louvain (Blondel et al.)	Greedy modularity optimization.	Traag et al.: up to 25% communities poorly connected.
Leiden (Traag et al., 2019)	Guarantees well-connected communities.	Sliwa et al.: detected 97 vs. 41 communities vs. Louvain.



Literature Review – Classical Algorithms

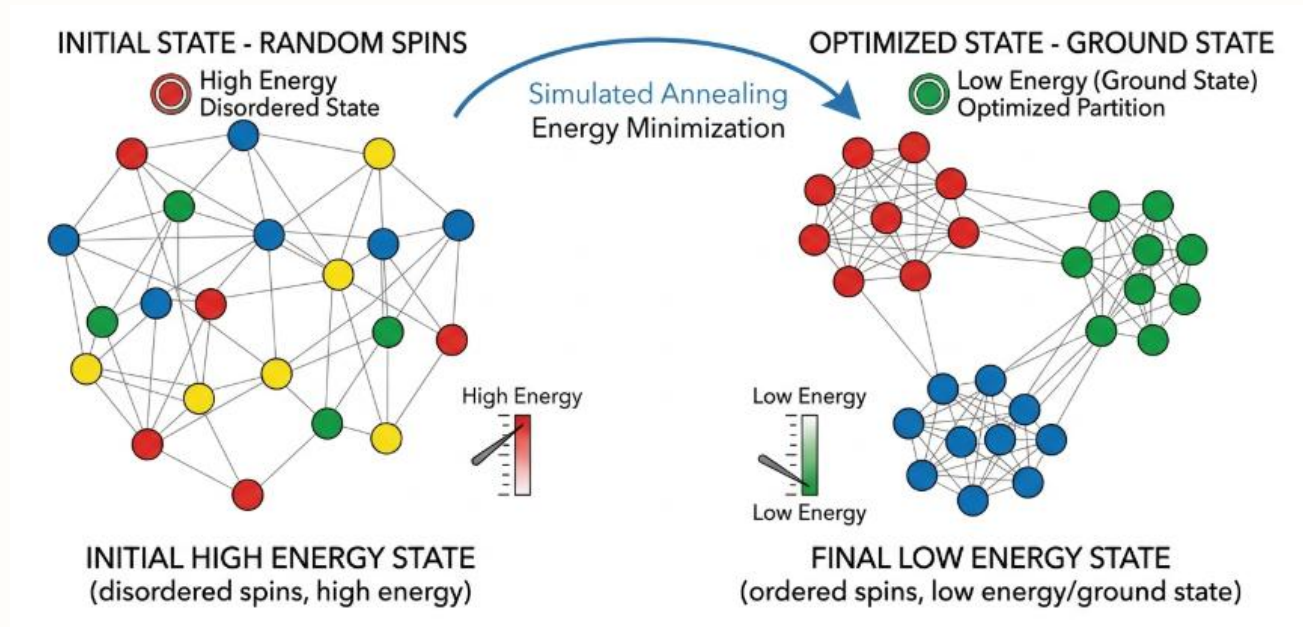
Random Walk-Based Approaches

<u>Algorithm</u>	<u>What It Does</u>	<u>Limitation</u>
Walktrap (Pons & Latapy)	Random walks trapped in dense regions.	Gates et al.: strong on flow-based networks.
Infomap (Rosvall & Bergstrom).	Minimizes map equation.	Yang et al.: linear time, top performer
MCL (van Dongen)	Expansion + inflation operators.	Papadopoulos et al.: pre-weighting improves NMI 0.4→0.8.



Literature Review – Classical Algorithms

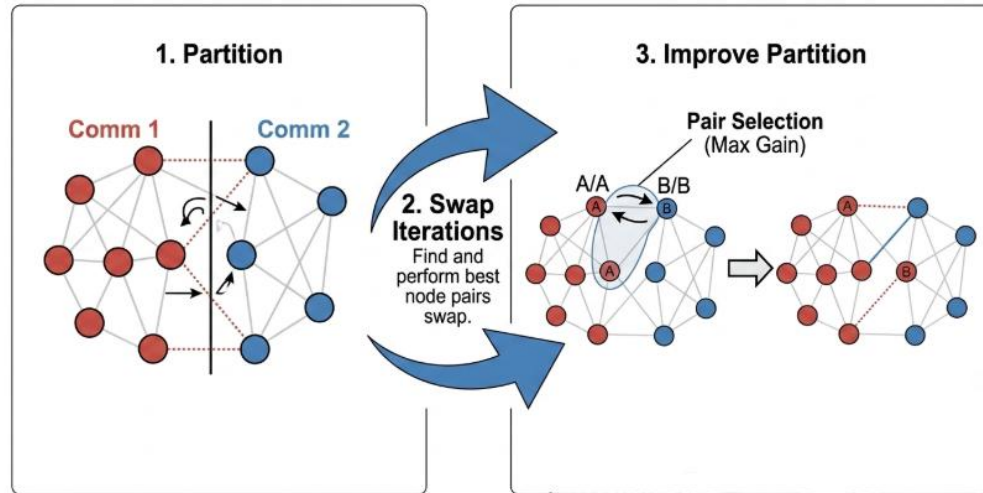
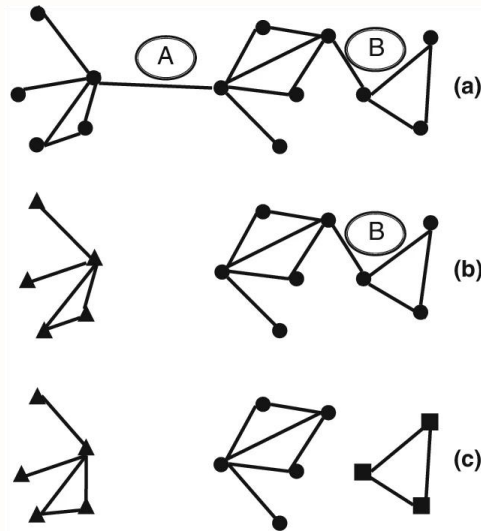
Statistical Physics-Based Algorithms - SpinGlass



Literature Review – Classical Algorithms

Divisive Approaches

<u>Algorithm</u>	<u>What It Does</u>	<u>Limitation</u>
Girvan-Newman	Removes highest betweenness edges	Gates et al.: $O(n^3)$, impractical beyond few thousand nodes.
Kernighan-Lin (1970)	Greedy bisection with local search	Li et al.: requires equal-sized partitions.



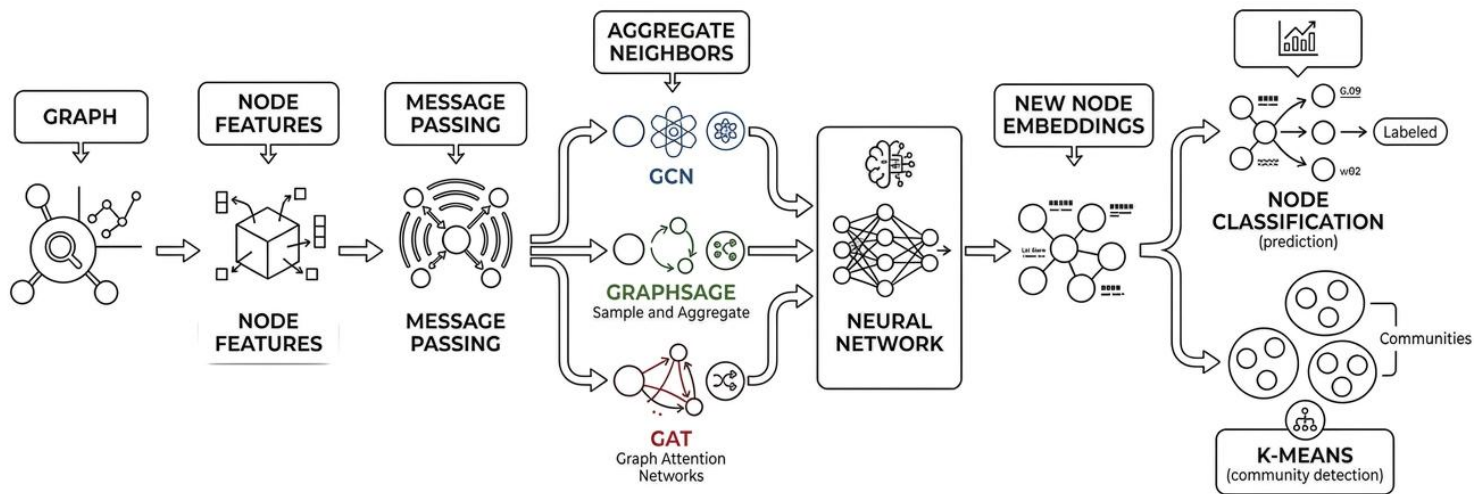
Literature Review – Classical Algorithms

Label Propagation Approaches

<u>Algorithm</u>	<u>What It Does</u>	<u>Limitation</u>
LPA (Raghavan et al.)	Nodes adopt most frequent neighbor label.	Gates et al.: fundamentally unstable.
FLPA	Asynchronous updates + memory	Yang et al.: outperforms LPA for $\mu \in [0.5, 0.7]$.
LPA-MNI (Li et al.)	Modularity + node importance.	Enhanced accuracy and robustness.
CLPA (Li et al.)	Constrained optimization.	Versatile for different network types.
FluidC (Parés et al.)	Communities as expanding fluids.	Anuar et al.: ARI ~0.9 on 500K nodes.

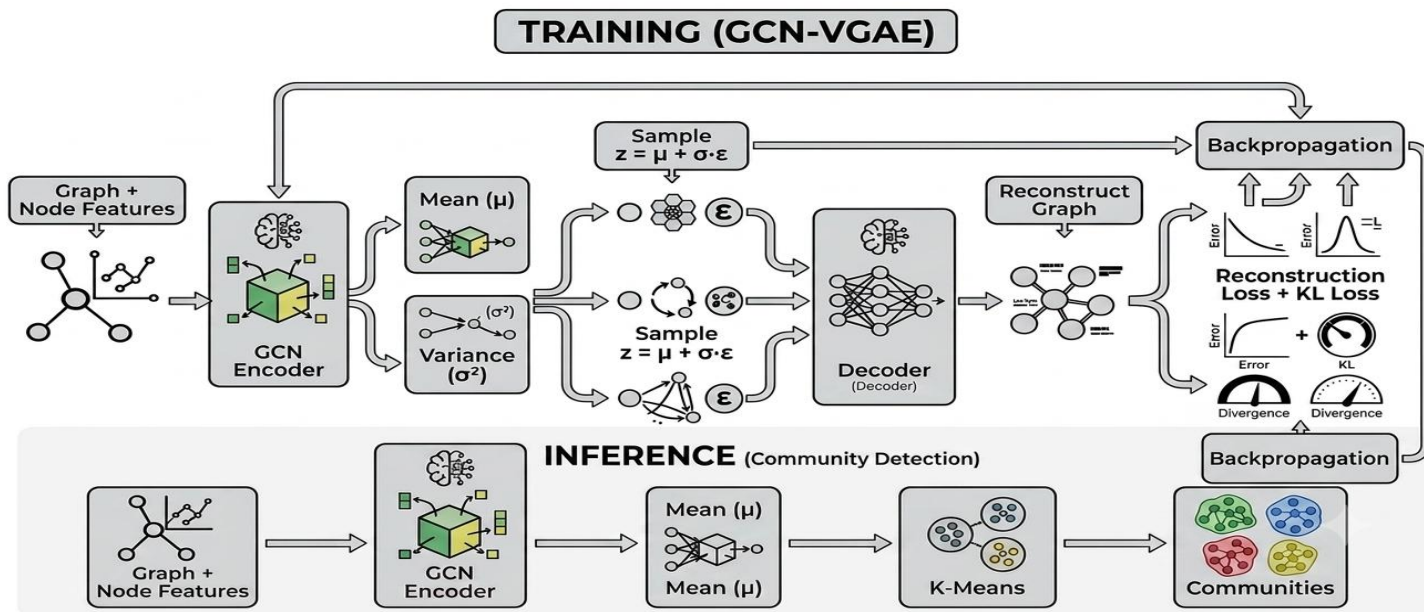
Literature Review – Modern Approaches

Graph Neural Networks (GCN – GraphSage – GAT)



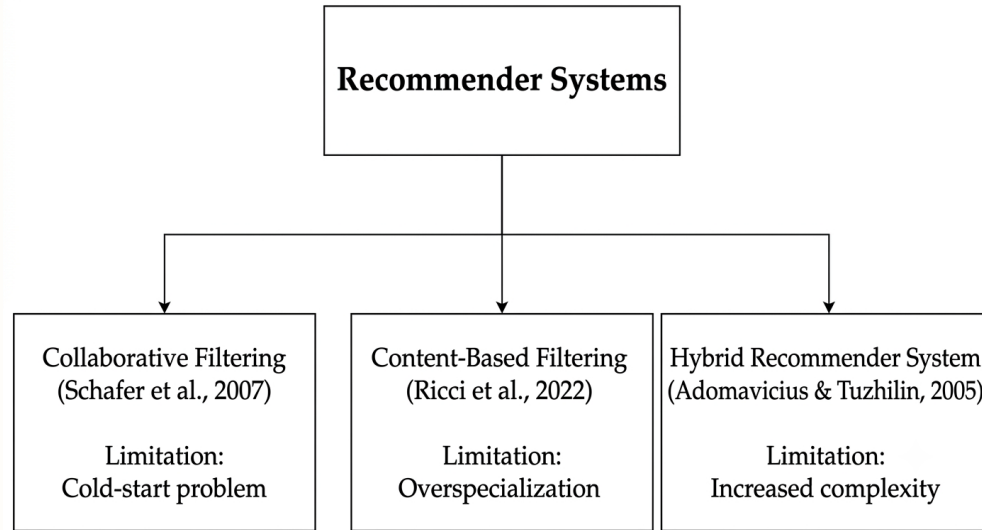
Literature Review – Modern Approaches

Graph Neural Networks (AE – VGAE)

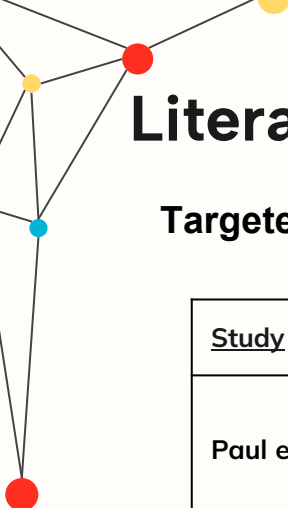


Literature Review – Recommender Systems

Three Classical Approaches



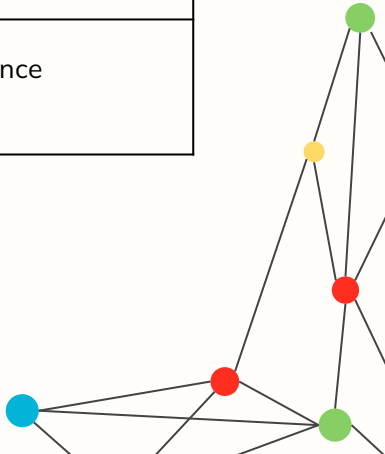
Challenges : Data Sparsity – Cold Start – Scalability – Interpretability.



Literature Review – Applications

Targeted Marketing and E-Commerce

<u>Study</u>	<u>Application</u>	<u>Results</u>
Paul et al. (2025)	Commerce Graph.	Customers with high centrality enable personalized promotions.
Nasirzonouzi (2025)	Customer segmentation.	Reveals complex interrelations K-means fails to capture.
Kanavos et al. (2024)	Twitter community analysis.	Community structures influence information dissemination.

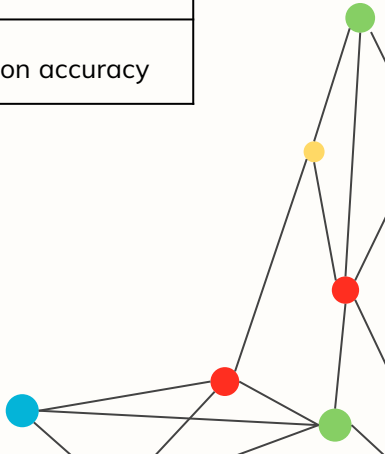




Literature Review – Applications

Recommender Systems Applications

<u>Study</u>	<u>Application</u>	<u>Results</u>
Saikalyan et al. (2024)	Local community detection	Effective cold-start on MovieLens and Facebook
Hu et al. (2024)	GGCRF framework	Addresses long-tail effect in recommendation
S.K.S. et al. (2025)	Textual semantics + GNNs	Improves community detection accuracy

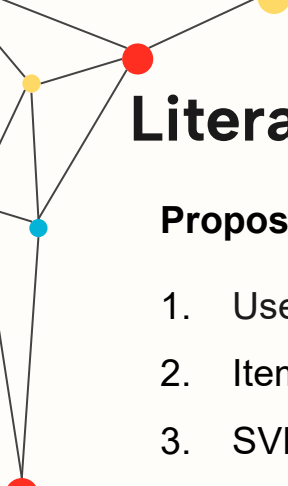


Literature Review – Applications

Standard Benchmark Datasets

<u>Dataset</u>	<u>Nodes</u>	<u>Edges</u>	<u>Communities</u>
Karate Club (Zachary, 1977)	34	78	2
Dolphins (Lusseau et al.)	62	159	2
Football (Girvan & Newman)	115	613	12
Polbooks (Krebs, 2004)	105	441	3
Primary School (Stehle et al., 2011)	236	125,000	10
Facebook (McAuley & Leskovec, 2012)	4,039	88,234	34
LastFM (McAuley & Leskovec, 2012)	28,281	92,752	18

LFR Synthetic Benchmark (Lancichinetti et al., 2008)



Literature Review – Applications

Proposed Hybrid Approaches

1. User-Based
2. Item-Based
3. SVD + Community
4. ALS + Community
5. BM25 + Community

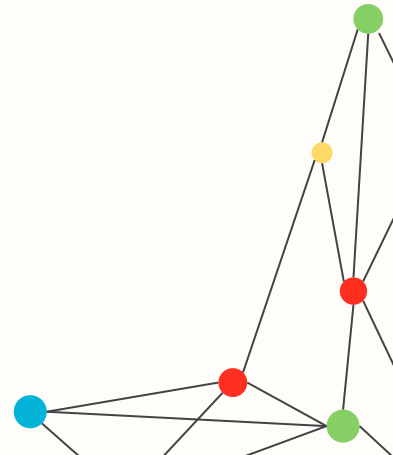
How They work

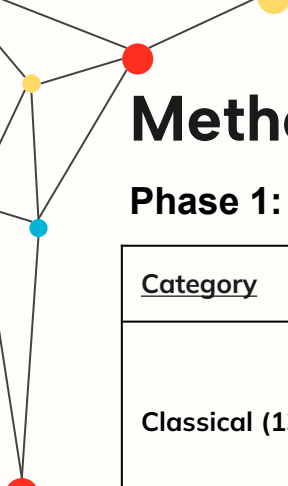
Community structure → Constrains candidate space, reduces sparsity.

Latent factors (SVD/ALS) → Captures fine-grained preferences.



BM25 → Leverages statistical co-purchasing patterns.





Methodology Overview

Phase 1: Community Detection Evaluation

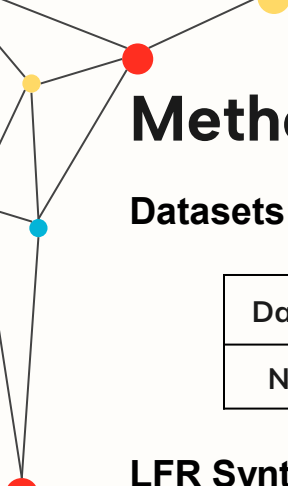
<u>Category</u>	<u>Algorithms</u>	<u>Datasets</u>	<u>Metrics</u>
Classical (13)	Louvain, Leiden, Walktrap, Infomap, LPA, FLPA, LPA-MNI, CLPA, FluidC, Girvan-Newman, Kernighan-Lin, SpinGlass, MCL	9 real-world + LFR synthetic (6 sizes)	NMI, ARI, Modularity
Deep Learning (4)	GCN, GAT, VGAE, Autoencoder	9 real-world + LFR synthetic	NMI, ARI, Modularity

Phase 2: Recommender System Integration

- Recommendation Approaches: User-Based, Item-Based, SVD+Community, ALS+Community, BM25+Community.
- Dataset: UCI Online Retail (541,909 transactions)
- Primary Metric: F1@5

Phase 3: Web Application





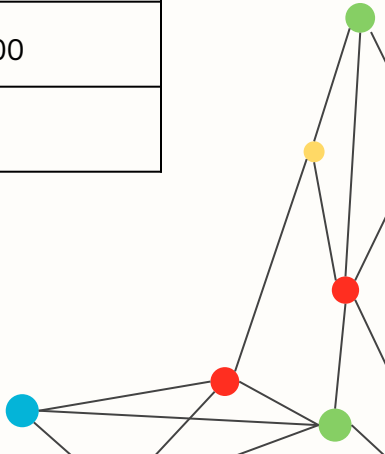
Methodology Overview – Datasets Used

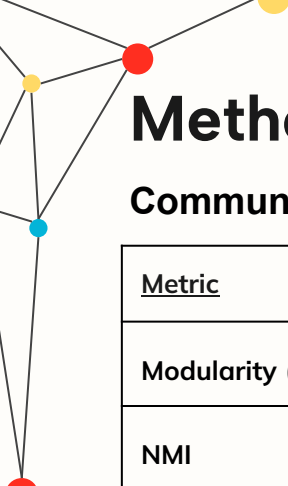
Datasets Summary

Dataset	Karate Club	Dolphins	Football	Polbooks	Primary School	Facebook	LastFM
Nodes	34	62	115	105	236	4,039	28,281

LFR Synthetic Networks

Parameter	Values
Network Size (n)	500, 1,000, 5,000, 10,000, 50,000, 100,000
Mixing Parameter (μ)	0.3, 0.5, 0.7





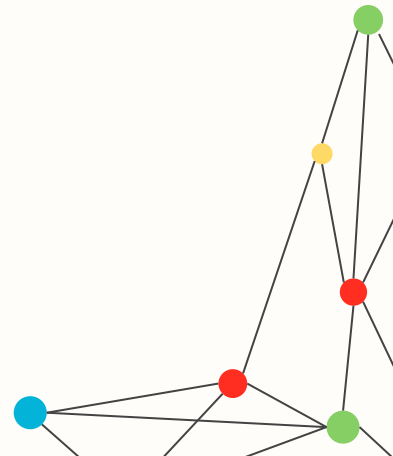
Methodology Overview – Evaluation Metrics

Community Detection Metrics

<u>Metric</u>	<u>Purpose</u>	<u>Range</u>
Modularity (Q)	Quality without ground truth	[-0.5, 1.0]
NMI	Similarity to ground truth	[0, 1]
ARI	Similarity adjusted for chance	[-1, 1]

Recommender System Metrics (@K)

- Precision@K : Relevant items in top-K / K
 - Recall@K : Relevant in top-K / Total relevant
 - F1@K : $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
- 



Results – Classical Algorithms

Community Detection Metrics - Modularity

Algorithm	Karate	Dolphins	Football	Polbooks	PS	VS13	VS15	FB	LastFM
Louvain	0.4151	0.5233	0.6046	0.5272	0.3708	0.3857	0.3415	0.813	0.8154
Leiden	0.4198	0.5276	0.6046	0.5269	0.3758	0.3857	0.3417	0.8179	0.8179
Walktrap	0.3532	0.4888	0.6029	0.507	0.3375	0.3794	0.3188	0.7519	0.7727
Infomap	0.402	0.5247	0.6005	0.5268	0.3758	0.3857	0.2849	0.754	0.7349
LPA	0.4033	0.5205	0.6032	0.5154	0.2687	0.3718	0.2654	0.7607	0.7976
FLPA	0.4151	0.5194	0.6046	0.5262	0.3589	0.3742	0.2654	0.7411	0.7584
LPA-MNI	0.39	0.5258	0.601	0.5248	0.3631	0.3741	0.2932	0.7494	0.767
CLPA	0.4174	0.5177	0.6046	0.5251	0.3754	0.3857	0.3415	0.8141	0.8147
FluidC	0.3758	0.4583	0.5671	0.4108	0.2519	0.27	0.2207	0.6733	0.5651
G-Newman	0.3488	0.3815	0.4549	0.4831	0.0008	0.0001	0	Slow	Slow
K-Lin	0.384	0.4197	0.5319	0.4297	0.2825	0.2594	0.2587	0.7174	0.7263
SpinGlass	0.4158	0.5285	0.6015	0.5259	0.3758	0.3855	0.3416	Slow	Slow
MCL	0.36	0.455	0.6005	0.5138	0.2819	0.1311	0.2659	0.637	0.4126

Results – Classical Algorithms

Community Detection Metrics - NMI

Algorithm	Karate	Dolphins	Football	Polbooks	PS	VS13	VS15	FB	LastFM
Louvain	0.6	0.4843	0.8903	0.5603	0.7805	0.8749	0.6329	0.2473	0.6278
Leiden	0.5878	0.5524	0.8903	0.5737	0.8475	0.8749	0.6408	0.2479	0.6254
Walktrap	0.4899	0.5816	0.8874	0.5427	0.6238	0.8662	0.5718	0.2967	0.583
Infomap	0.5684	0.5373	0.9242	0.5369	0.8475	0.8749	0.598	0.2976	0.5277
LPA	0.5056	0.6596	0.8866	0.5497	0.4169	0.8345	0.4284	0.2748	0.6294
FLPA	0.6	0.6256	0.8903	0.5537	0.6307	0.8665	0.4284	0.276	0.5791
LPA-MNI	0.5111	0.588	0.9269	0.5259	0.9281	0.8662	0.5703	0.2733	0.583
CLPA	0.5925	0.5345	0.8903	0.5427	0.8072	0.8749	0.6329	0.2467	0.6307
FluidC	0.4639	0.4807	0.8443	0.4003	0.8993	0.696	0.6848	0.2238	0.457
G-Newman	0.6804	0.9085	0.4872	0.5754	0.0339	0.0435	0.0298	Slow	Slow
K-Lin	0.493	0.3819	0.7791	0.395	0.8071	0.6862	0.6169	0.068	0.3771
SpinGlass	0.5622	0.5565	0.9065	0.4781	0.8475	0.8784	0.6448	Slow	Slow
MCL	0.7324	0.4386	0.9242	0.5209	0.4237	0.4425	0.4385	0.2885	0.4616

Results – Classical Algorithms

Community Detection Metrics - ARI

Algorithm	Karate	Dolphins	Football	Polbooks	PS	VS13	VS15	FB	LastFM
Louvain	0.5089	0.3472	0.8069	0.6567	0.5032	0.8986	0.5292	0.1003	0.573
Leiden	0.4646	0.3809	0.8069	0.6752	0.6061	0.8986	0.5355	0.0973	0.5452
Walktrap	0.3207	0.4662	0.8154	0.6534	0.309	0.8722	0.4426	0.1134	0.482
Infomap	0.5906	0.3659	0.8967	0.6463	0.6061	0.8986	0.3606	0.0277	0.2334
LPA	0.4588	0.4815	0.7913	0.6667	0.1666	0.8427	0.252	0.0355	0.5527
FLPA	0.5089	0.4505	0.8069	0.6649	0.3396	0.8611	0.252	0.0414	0.4449
LPA-MNI	0.4809	0.4293	0.8893	0.6116	0.7994	0.8605	0.3389	0.037	0.4591
CLPA	0.4831	0.3821	0.8069	0.6487	0.5581	0.8986	0.5292	0.1068	0.5639
FluidC	0.3285	0.2799	0.6938	0.185	0.8267	0.4569	0.4626	0.0209	0.1026
G-Newman	0.7212	0.9175	0.2045	0.6795	0.0007	0.005	0.0037	Slow	Slow
K-Lin	0.3693	0.1515	0.6163	0.202	0.7011	0.4513	0.4808	0.052	0.2994
SpinGlass	0.4313	0.3792	0.8558	0.4534	0.6061	0.8986	0.5403	Slow	Slow
MCL	0.7717	0.211	0.8967	0.5936	0.1718	0.202	0.2575	0.0182	0.0407

Results – Deep Learning Methods

Results - Deep Learning Methods - NMI

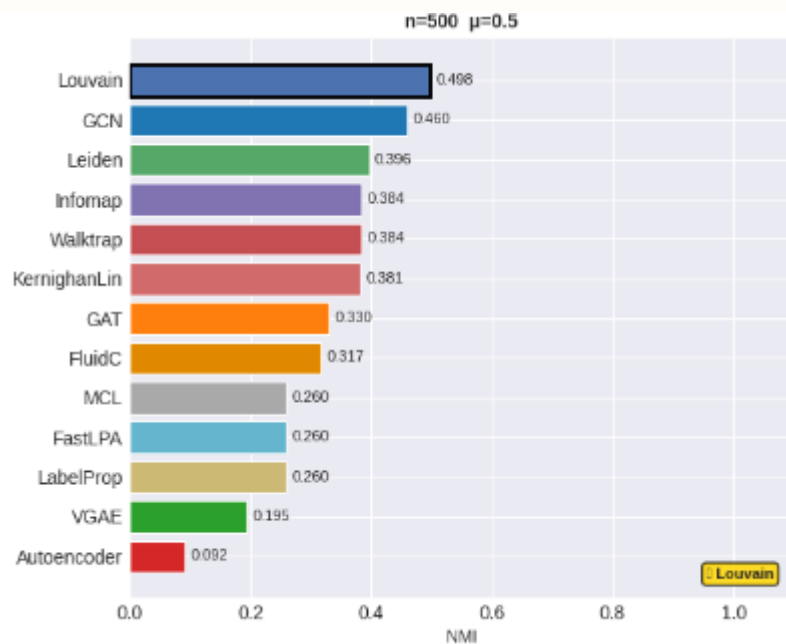
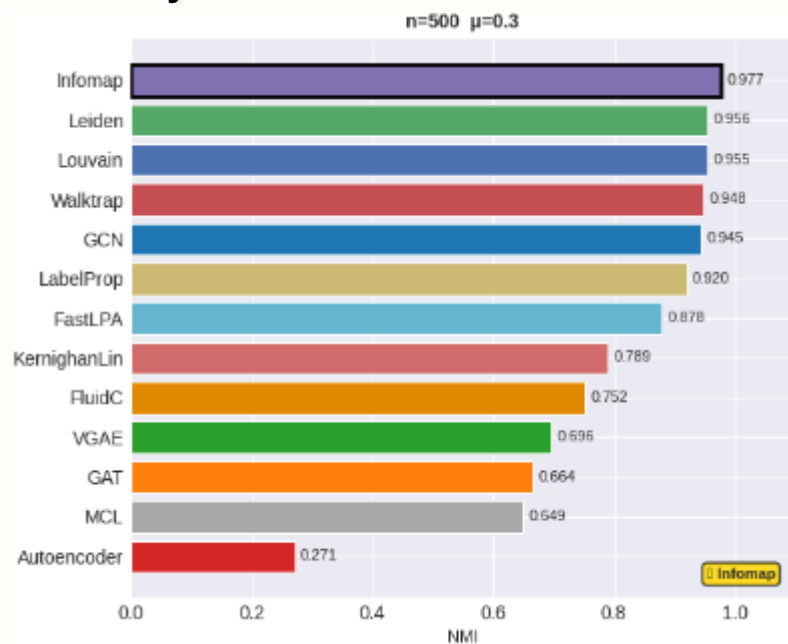
Algorithm	Karate	Dolphins	Football	Polbooks	PS	VS13	VS15	FB	LastFM
GCN	0.8372	0.8888	0.925	0.5745	0.1421	0.7871	0.6342	0.0486	0.0313
GAT	0.5756	0.8165	0.9195	0.4592	0.1119	0.8214	0.6343	0.0441	0.0295
VGAE	0.7324	1	0.9231	0.445	0.1263	0.7886	0.6095	0.0348	0.0258
Autoencoder	0.093	0.0279	0.8973	0.1635	0.1177	0.6207	0.567	0.0525	0.0296

Results - Deep Learning Methods - ARI

Algorithm	Karate	Dolphins	Football	Polbooks	PS	VS13	VS15	FB	LastFM
GCN	0.8823	0.9348	0.8995	0.6745	0.0272	0.6808	0.4339	0.0614	0.0073
GAT	0.6682	0.8725	0.8955	0.4903	0.0129	0.7574	0.4490	0.0528	0.0064
VGAE	0.7717	1.0000	0.8905	0.4953	0.0234	0.7252	0.4248	0.0278	0.0049
Autoencoder	0.0072	0.0578	0.8358	0.0181	0.0144	0.3707	0.3719	0.0622	0.0061

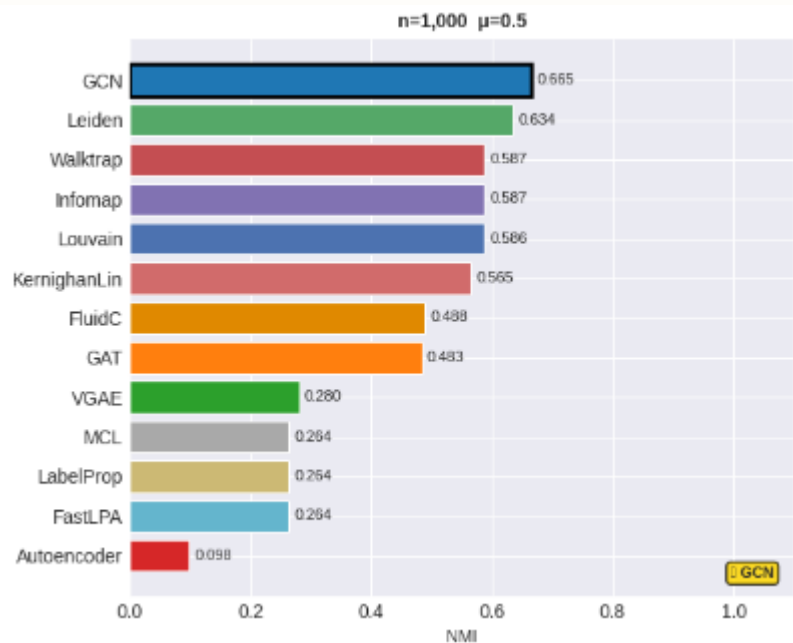
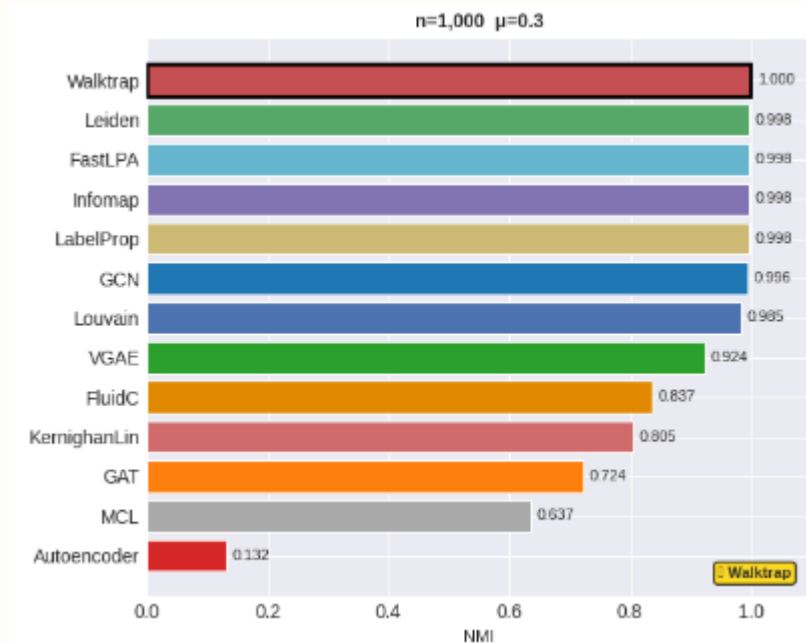
Results – LFR Benchmark

Community Detection Metrics - NMI



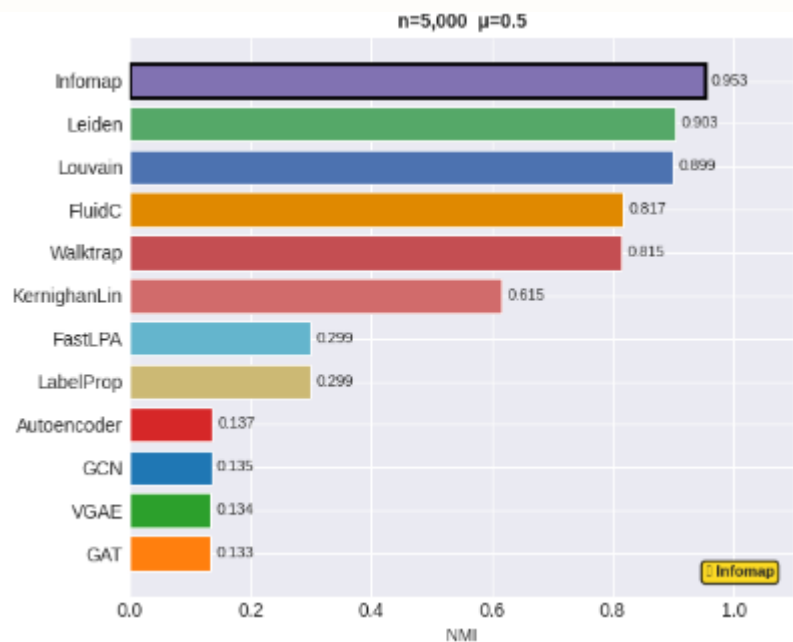
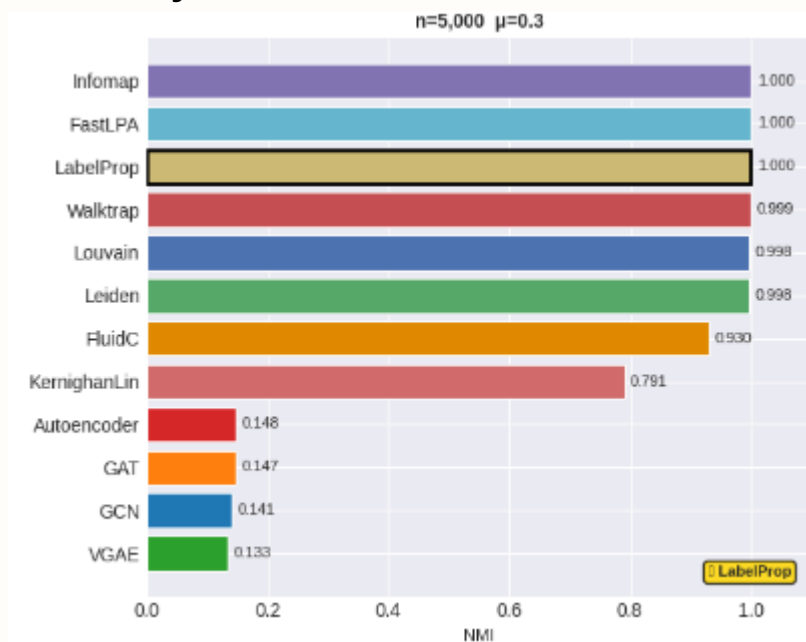
Results – LFR Benchmark

Community Detection Metrics - NMI



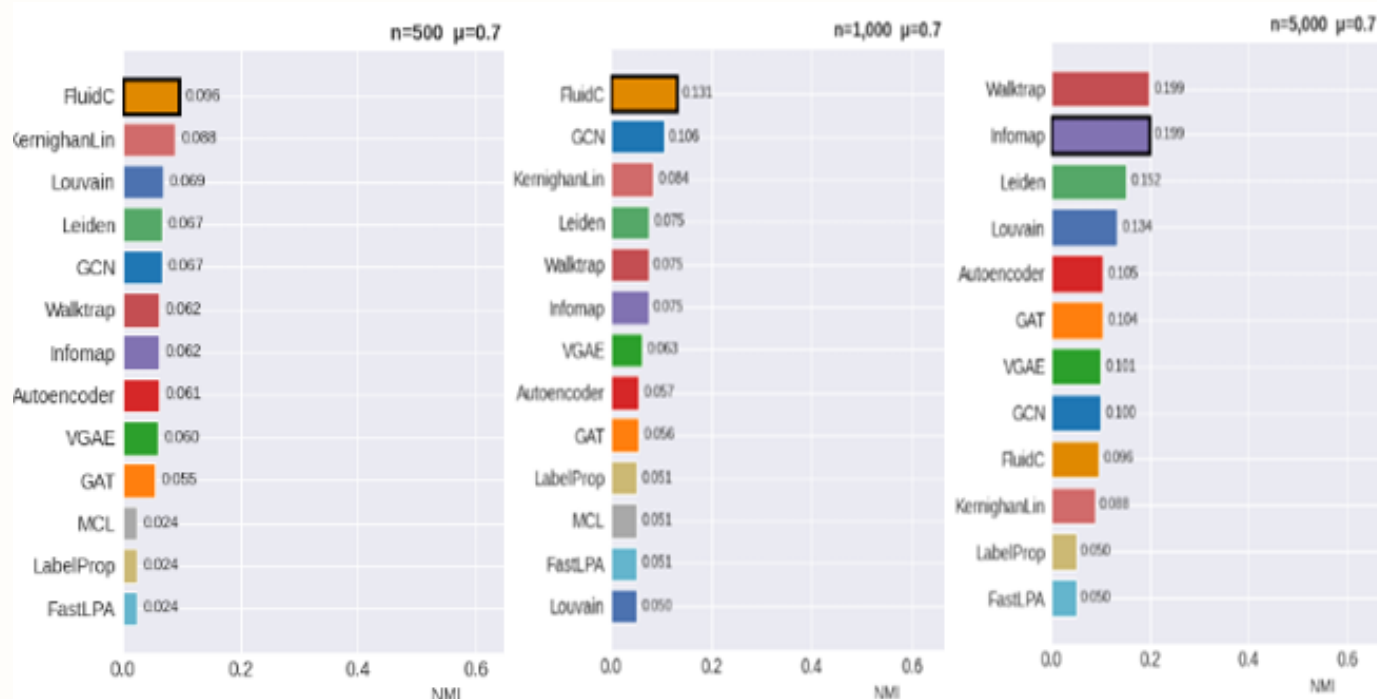
Results – LFR Benchmark

Community Detection Metrics - NMI



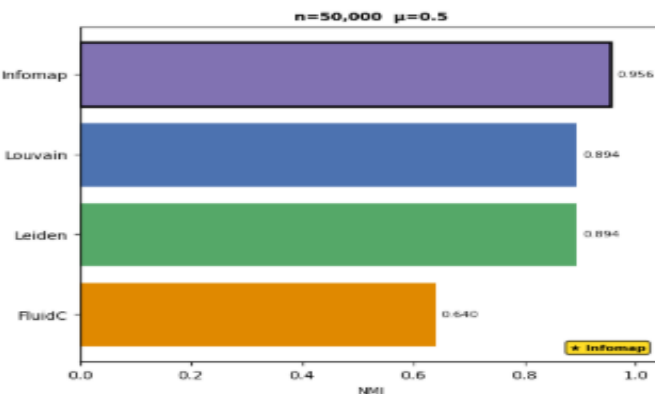
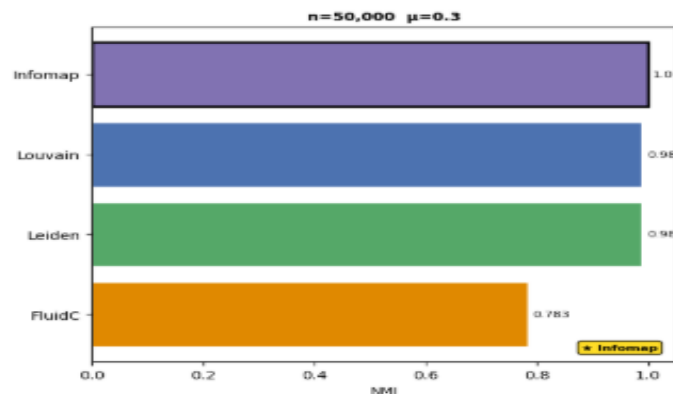
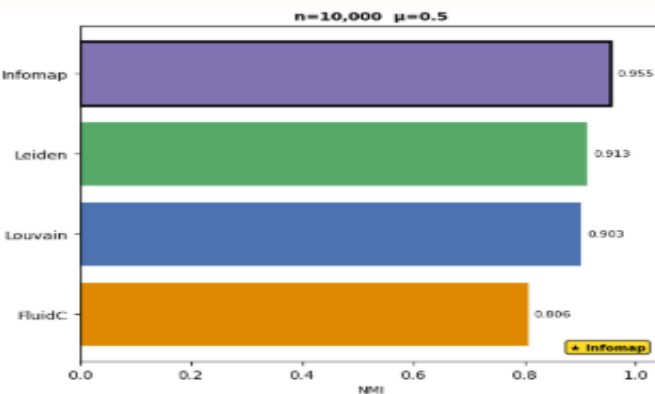
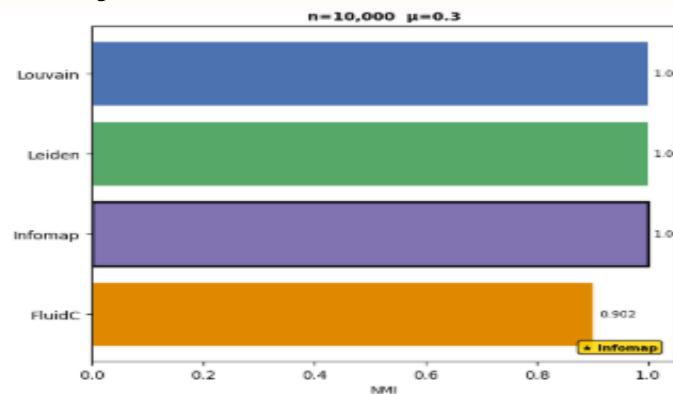
Results – LFR Benchmark

Community Detection Metrics - NMI



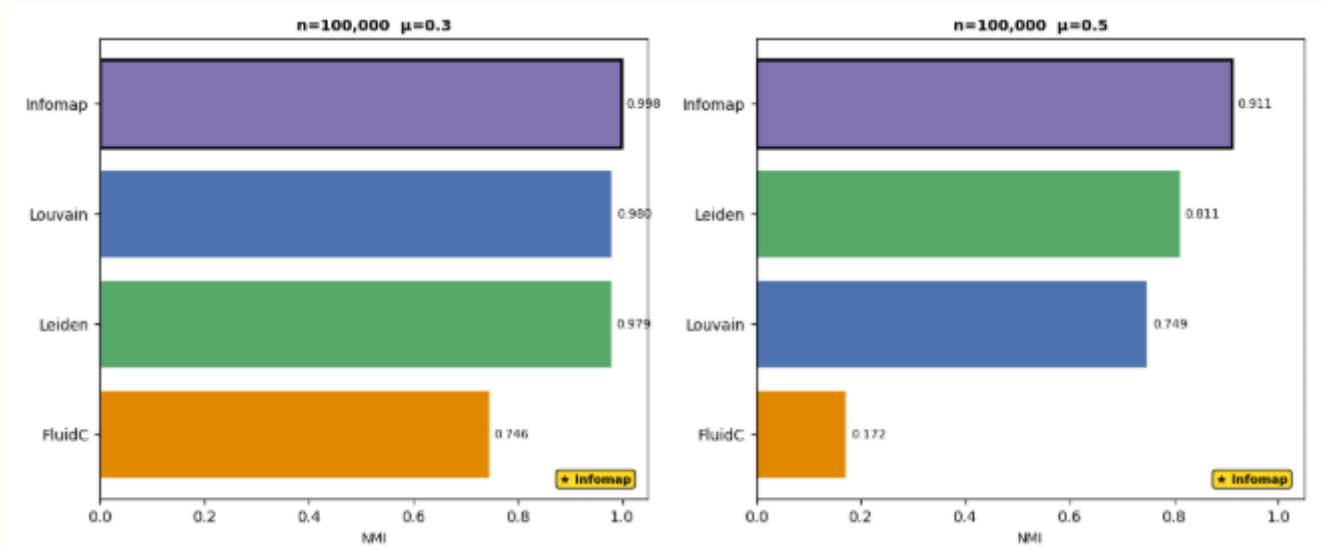
Results – LFR Benchmark

Community Detection Metrics - NMI



Results – LFR Benchmark

Community Detection Metrics - NMI



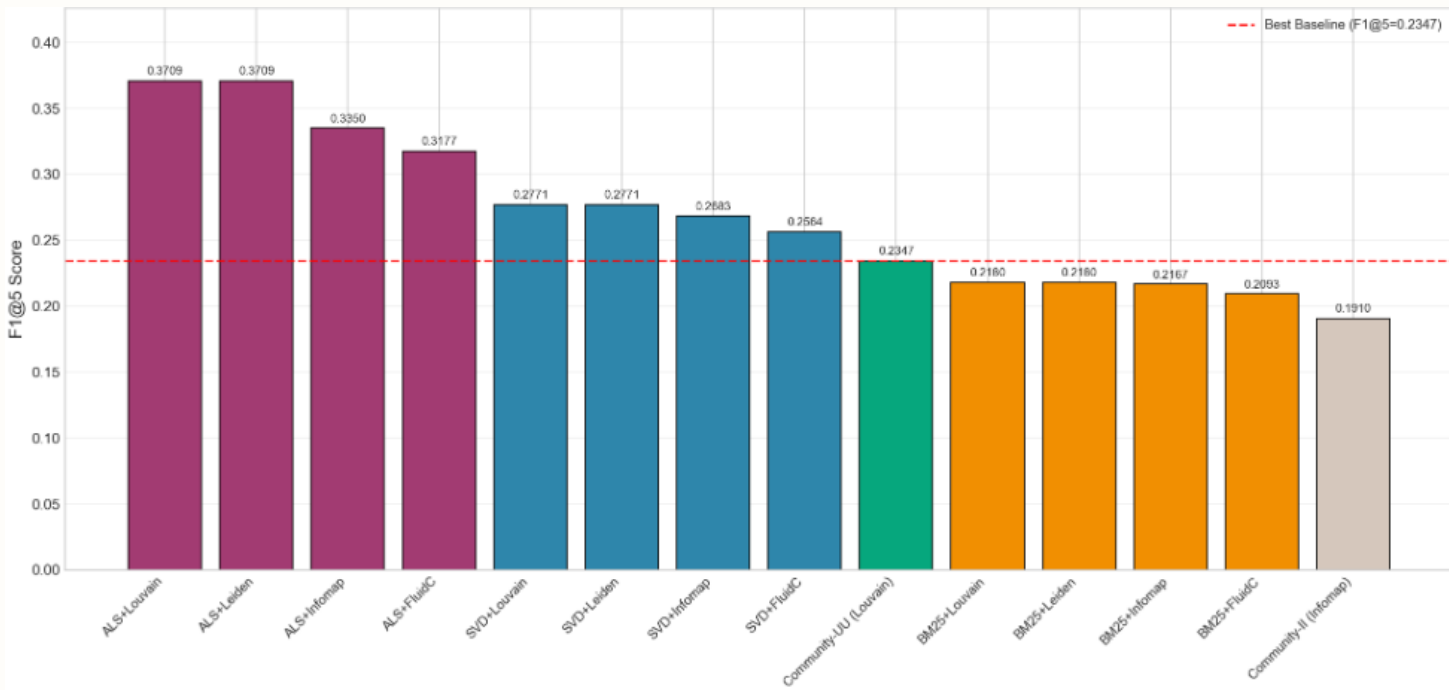
Results – Recommender System Integration

Hyperparameter Tuning Results

<u>Model</u>	<u>Default Config</u>	<u>Default F1@5</u>	<u>Tuned Config</u>	<u>Tuned F1@5</u>
ALS	factors=20, reg=0.1, $\alpha=40$	0.149	factors=100, reg=0.1, $\alpha=10$	0.371
BM25	k1=1.5, b=0.75	0.178	k1=3.0, b=1.0	0.211
SVD	n_components=50	0.277	n_components=50	0.277

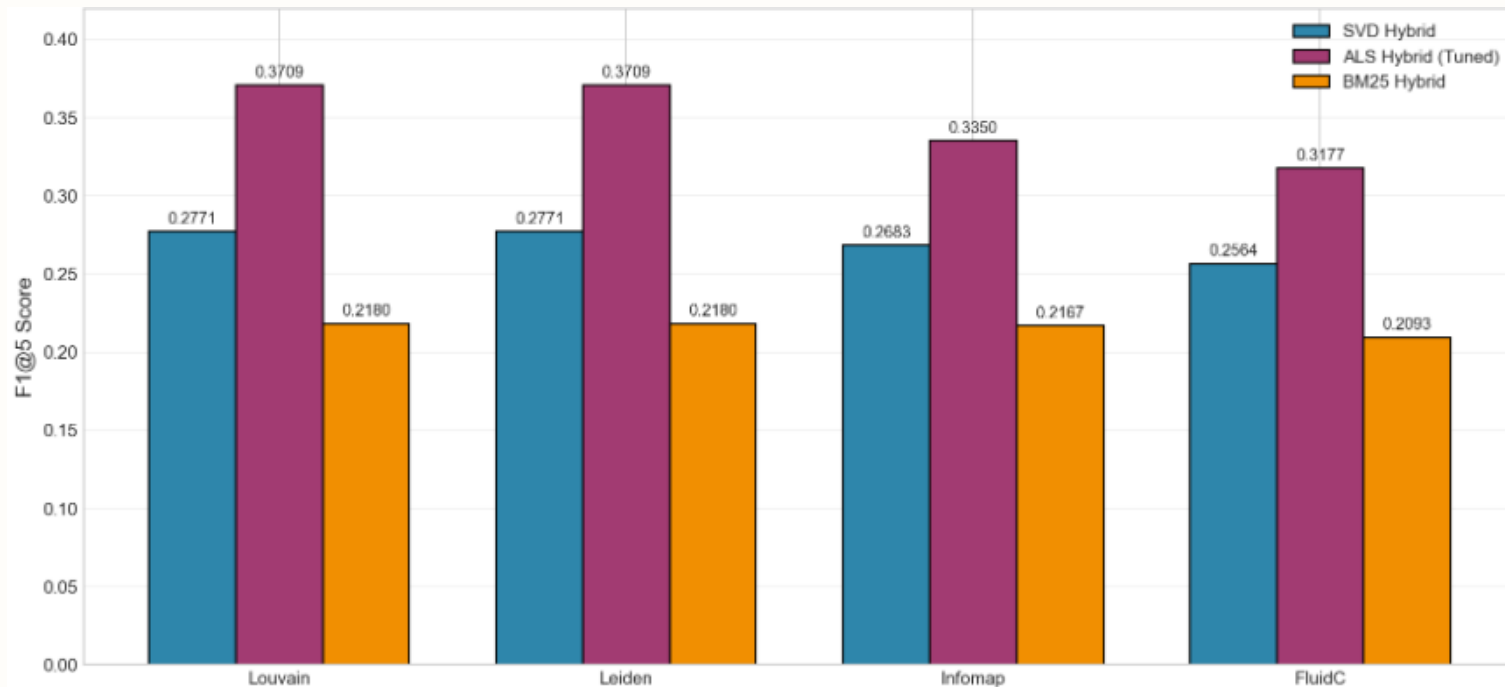
Results – Recommender System Integration

Performance Summary



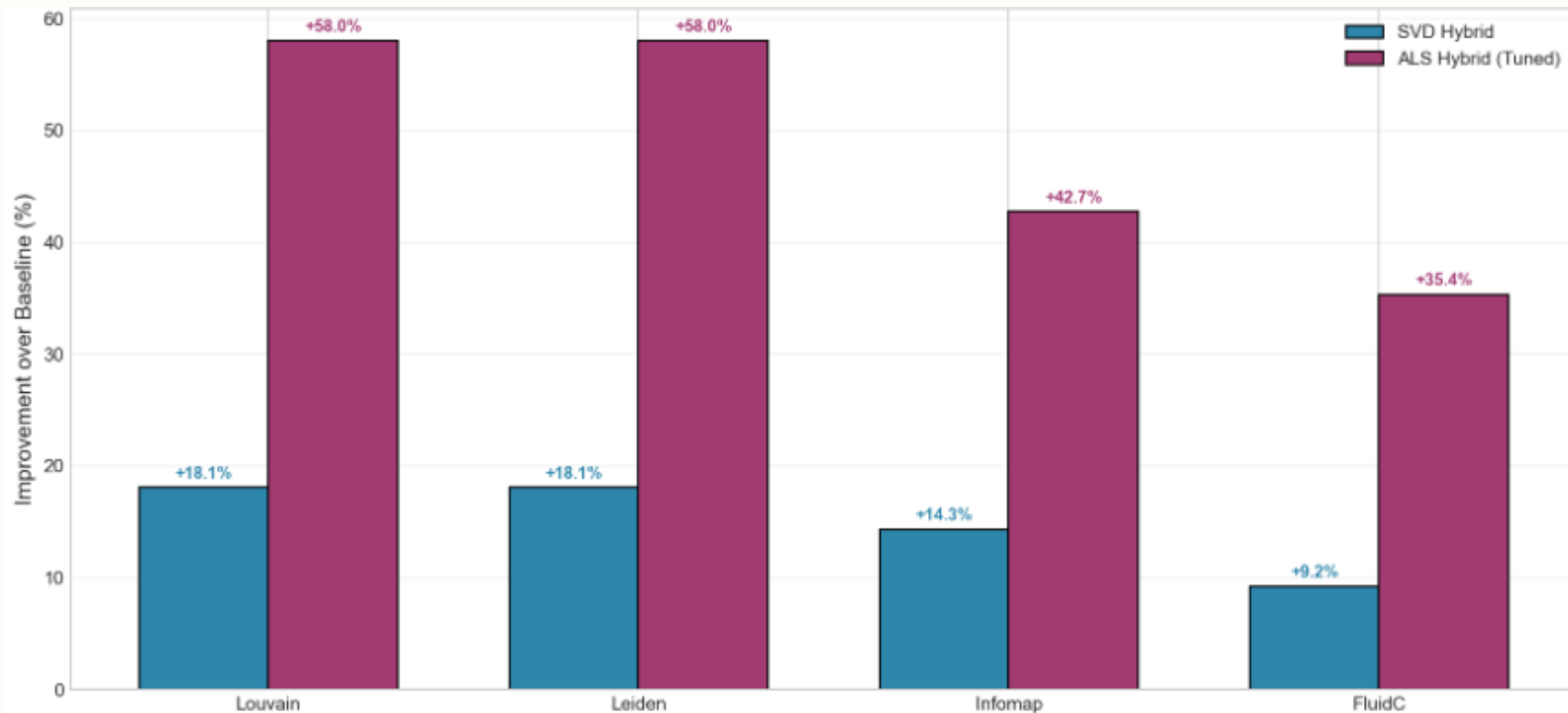
Results – Recommender System Integration

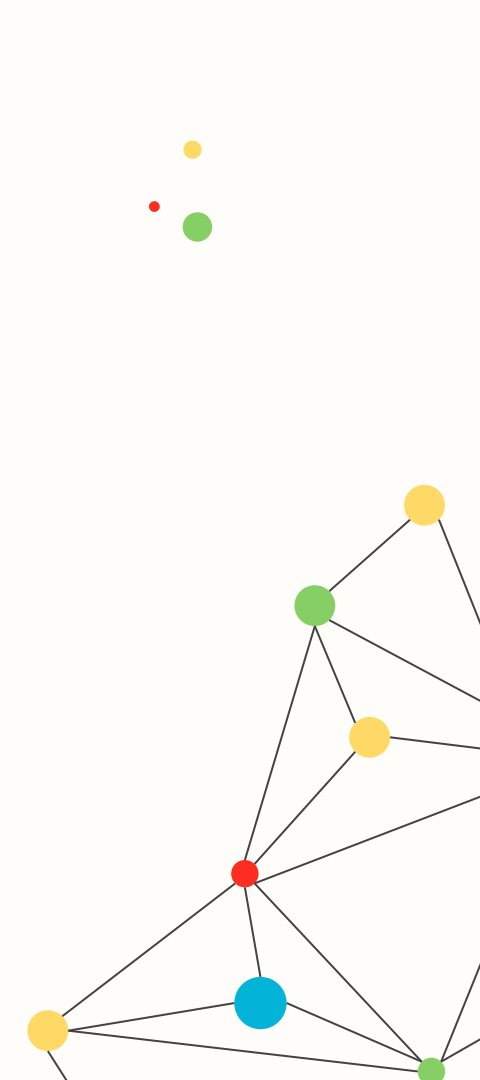
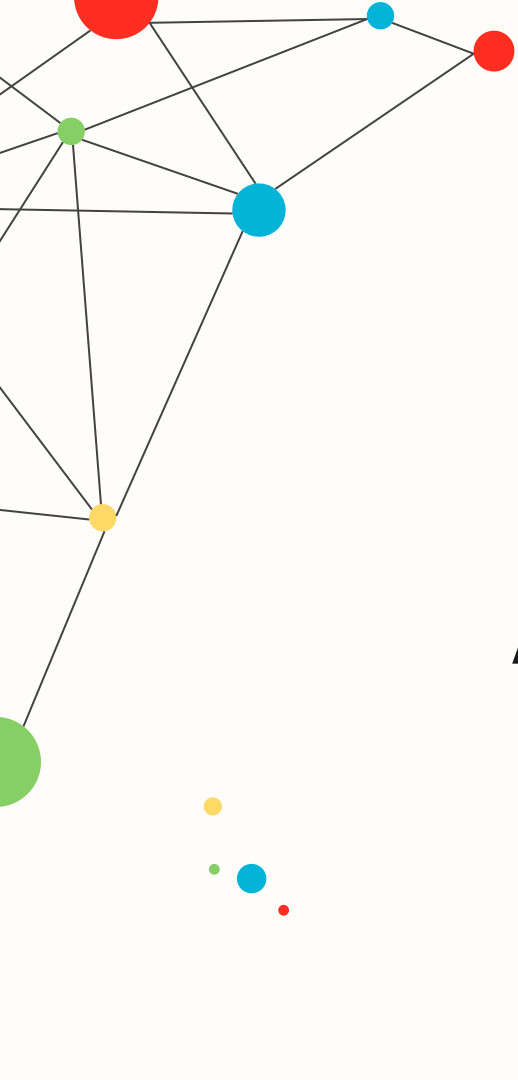
Performance Summary



Results – Recommender System Integration

Performance Summary





WEB APPLICATION



Discussion

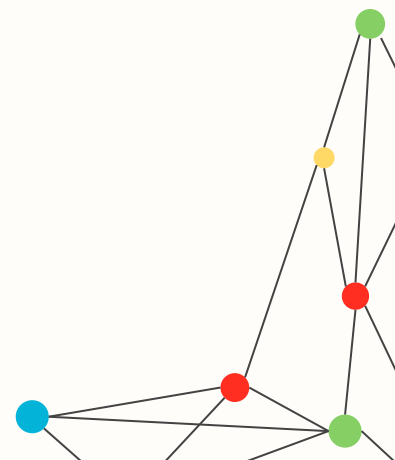
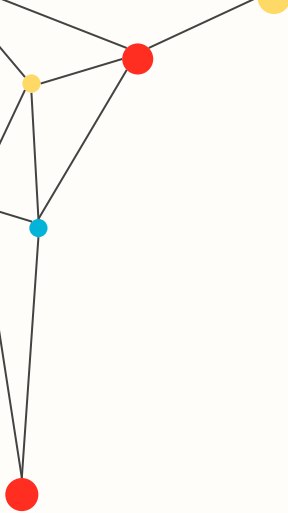
Research Question: *How do different community detection approaches compare in effectiveness and practical utility for recommender systems?*

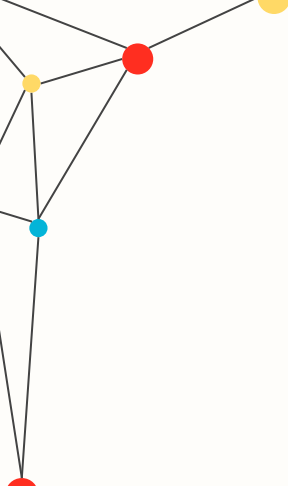
Classical algorithms (Louvain, Leiden, Infomap) remain the most effective and practical choice

Strengths	Details
High structural quality	$Q > 0.81$ on large networks
Excellent scalability	Near-linear time
Substantial RS improvements	+58% when combined with tuned ALS



Conclusion





**Thank you for your
attention**

